



Regression Analysis

Statistical method used to examine the relationship b/w one dependent variable and one or more independent variable.

How y changes when $x_1, \dots, x_n$ are altered.

Predict the y as well.

Flow :

1. Define the research question.
2. Collect and prepare data.
3. Visualize data.
4. Check assumptions.
5. Fit the linear regression model.
- ➔ 6. Interpret the model. (Regression analysis is done)
7. Validate the model.
8. Report results.

Agenda : —

What is statistics connection?

Why is regression analysis required?

Why ML problems are statistical inference problem?

Sample	Population.
n	N .
Avg. salary in India	
1000	1.4B
$\bar{x}$	μ
	estimate.

Supervised ML

- Regression.
- Classification.

Regression.

	x_1 cgpa	x_2 ig	$y \rightarrow$ predict. lpa
$n=100$ {	-	-	-
	-	-	-
	-	-	-

This is ML problem.

$$lpa = f(cgpa, ig)$$

input $\longrightarrow$ output.
(predict)

$n=100$ student are given but we need to find relation (i.e. mathematical fn) for all N .

$$lpa = f(cgpa, ig) + \epsilon$$

$\uparrow$
error (irreducible)

In real world, there are conditions that cannot be considered mathematically. viable, or biases.

$$lpa = \underbrace{f(cgpa, ig)}_{\substack{\text{parametric} \quad \text{non-parametric}}} + \epsilon$$

eg.) $lr \rightarrow \underbrace{\beta_0}_{\uparrow} + \underbrace{\beta_1}_{\uparrow} cgpa + \underbrace{\beta_2}_{\uparrow} ig$ eg.) Decision Tree

$$\text{for } \rightarrow L_{pa} = \beta_0 + \beta_1 c_{gpa} + \beta_2 i_g.$$

$$\text{Coef } \rightarrow b_0, b_1, b_2 \neq \underset{\substack{\text{(not} \\ \text{exact)}}}{\beta_0, \beta_1, \beta_2}.$$

samples population's

What we get due to n not N .

$$\begin{array}{c} f'(c_{gpa}, i_g) \\ \uparrow \\ \text{sample} \end{array}$$

$$\text{error} \rightarrow f() - f'()$$

This is called irreducible error.

Now,

$$L_{pa} = f'(c_{gpa}, i_g) + \underbrace{\text{irreducible error}}_{\substack{\uparrow \\ \text{Task of ML eng.}}} + \varepsilon$$

$$f'() \approx f()$$

estimated
relationship
of x, y based
on given data
(samples)

True relationship
of x & y for
population data.

Inferences vs. Prediction [Why regression analysis is required?]

Diff. b/w inference & prediction.

prediction ↓

cgpa	iq	lpa
-	-	-
-	-	-
-	-	-

↓

lr ($\beta_0, \beta_1, \beta_2$)

↓

$$lpa = \beta_0 + \beta_1 \text{cgpa} + \beta_2 \text{iq}$$

↓ Ans.

Prediction value.

inference : study of relationship.

$lpa \rightarrow \text{cgpa, iq}$

$y \quad \quad \quad x \quad \quad \text{relationship.}$

$y \rightarrow \text{cgpa}$
 $y \rightarrow \text{iq}$ } relationship

∴ We need regression analysis.

Coefficient (m, b) significance

and also the relationship b/w x & y.

eg. Blackbox model. (where we have prediction but not inference)

Linear regression $\rightarrow$ very good at inference, but weak prediction.

generally $\rightarrow$ good inference so weak prediction & vice-versa.

Whenever we have inference, we need regression analysis.

Statsmodel Linear Regression.:

$X \rightarrow Y$: Is there a relationship?

Is that linear?

How strong? How strong each component of $X(x_1, x_2, \dots, x_n)$ with Y ?

OLS Regression Results

Dep. Variable:	Sales	R-squared:	0.897
Model:	OLS	Adj. R-squared:	0.896
Method:	Least Squares	F-statistic:	570.3
Date:	Sun, 15 Jun 2025	Prob (F-statistic):	1.58e-96
Time:	23:35:26	Log-Likelihood:	-386.18
No. Observations:	200	AIC:	780.4
Df Residuals:	196	BIC:	793.6
Df Model:	3		
Covariance Type:	nonrobust		

Goodness
of fit
i.e relation
between X & Y .

	coef	std err	t	P> t	[0.025	0.975]
const (β_0)	2.9389	0.312	9.422	0.000	2.324	3.554
TV (β_1)	0.0458	0.001	32.809	0.000	0.043	0.049
Radio (β_2)	0.1885	0.009	21.893	0.000	0.172	0.206
Newspaper (β_3)	-0.0010	0.006	-0.177	0.860	-0.013	0.011

Shows relation
between $Y \rightarrow TV(x_1)$
 $Y \rightarrow Radio(x_2)$
 $Y \rightarrow Newspaper(x_3)$

Omnibus:	60.414	Durbin-Watson:	2.084
Prob(Omnibus):	0.000	Jarque-Bera (JB):	151.241
Skew:	-1.327	Prob(JB):	1.44e-33
Kurtosis:	6.332	Cond. No.	454.

$\rightarrow$ Testing the assumptions of L.R.

(Point 4: Check assumption)

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Code ↗

```
import pandas as pd
import statsmodels.api as sm

# Load the dataset
url = "https://raw.githubusercontent.com/justmarkham/scikit-learn-videos/master/data/Advertising.csv"
data = pd.read_csv(url, index_col=0)

# Define the independent variables (add a constant for the intercept)
X = data[['TV', 'Radio', 'Newspaper']]
X = sm.add_constant(X)

# Define the dependent variable
y = data['Sales']

# Fit the model using the independent and dependent variables
model = sm.OLS(y, X).fit()

# Print the summary of the model
print(model.summary())
```

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F-Statistic : Is there any relationship b/w x and y.
 F-Statistic (prob.) : How strong is relation?

Hypothesis test $\rightarrow$ F-test for overall significance (ANOVA)

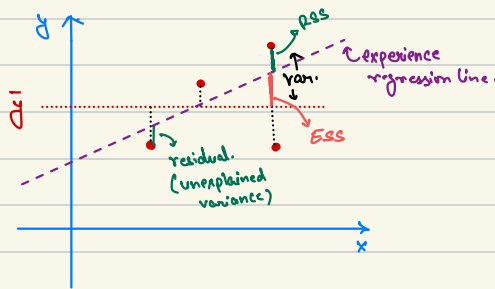
TSS, RSS, and ESS :

TSS $\rightarrow$ Total Sum of Squares.

RSS $\rightarrow$ Residual Sum of Squares.

ESS $\rightarrow$ Explained Sum of Squares.

experience | salary.



$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

• Overall variance in data.

$$TSS \rightarrow ESS + RSS$$

↑ explained ↑ unexplained (E)

↓ reducible ↓ irreducible.

• Variance around regression line.

Degree of Freedom:

Degree of Freedom

28 April 2023 07:00

In linear regression, the total degrees of freedom (df_{total}) represent the total number of data points minus 1. It represents the overall variability in the dataset that can be attributed to both the model and the residuals.

For a linear regression with n data points (observations), the total degrees of freedom can be calculated as:
 $df_{total} = n - 1$
 where: n is the number of data points (observations) in the dataset

The total degrees of freedom in linear regression is divided into two components:

1. Degrees of freedom for the model (df_{model}): This is equal to the number of independent variables in the model (k).
2. Degrees of freedom for the residuals ($df_{residuals}$):
 The degrees of freedom for the residuals indicate the number of independent pieces of information that are available for estimating the variability in the residuals (errors) after fitting the regression model.
 This is equal to the number of data points (n) minus the number of estimated parameters, including the intercept ($k+1$).

The sum of the degrees of freedom for the model and the degrees of freedom for the residuals is equal to the total degrees of freedom:
 $df_{total} = df_{model} + df_{residuals}$

$$df_{total} = df_{model} + df_{residual}$$

(k) (n-k-1)

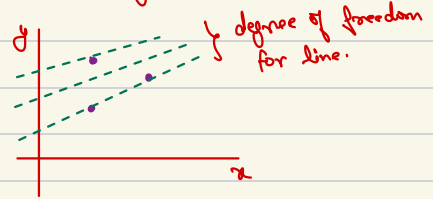
$n \rightarrow$ no. of rows.

$k \rightarrow$ total independent vars. (i.e., $x_1, x_2, \dots, x_n$)
 (here, $3 \rightarrow$ TV, Radio, News)

$$df_{total} = k + n - k - 1$$

$$df_{total} = n - 1$$

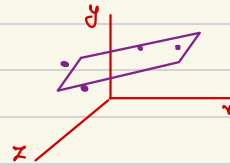
$$\ln(L) \rightarrow x/y$$



regression line $\rightarrow$ min. data points required = 3

Now freedom $df = 1$

if $n = 3, k = 1 \rightarrow df_{residual} = 1$



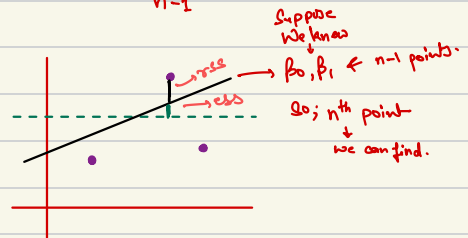
min data point req = 4

$df_{model}(k) = 2$

$df_{residual} = n - k - 1$
 $= 4 - 2 - 1$
 $= 1$

$df_{residual} \rightarrow$ The freedom to move.

$df_{total} \rightarrow \beta_0, \beta_1$
 $n - 1$



F-statistic & Prob.(F-Statistic) :

F-statistic model tells does X, Y have any relationship or not.
independent $\rightarrow$ dependent.

exp. | sal.
Does exp. required in prediction of salary | or Does exp. required in predicting salary or we can just take mean (sal.).

F-Test for overall significance :

Steps :-

	exp X	sal. Y
$n=100$	-	-
	-	-
	-	-

$$\underset{\substack{\uparrow \\ \text{sal.}}}{y} = \beta_0 + \beta_1 \underset{\substack{\uparrow \\ \text{exp.}}}{x}$$

$$\begin{aligned} \text{Take } H_0 : & \beta_1 = 0 \\ H_1 : & \beta_1 \neq 0 \end{aligned}$$

(for multiple inp $\rightarrow \beta_1, \beta_2 \dots \beta_n = 0$)

(At least one is not equal to zero)

$$MSR = \frac{ESS}{K \leftarrow df_{\text{model}}}$$

$$MSE = \frac{RSS}{n-k-1 \leftarrow df_{\text{residual}}}$$

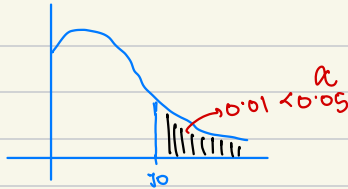
$$F\text{-Statistic} = \frac{\frac{\chi^2_{\text{Distribution}}}{df_1}}{\frac{\chi^2_{\text{Distribution}}}{df_2}}$$

$$F\text{-Statistic} = \frac{MSR}{MSE} \rightarrow \begin{aligned} & \text{Means Square Regression} \\ & \text{MSE} \rightarrow \text{Mean Squared Error.} \end{aligned}$$

$$MSR = \frac{TSS - RSS}{K}$$

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{n - k - 1}$$

Calculate p-value for F-statistic.



Reject H_0 .

Explanations :-

$$F\text{-Stat} = \frac{\frac{ESS}{K} \rightarrow \text{avg explained var per df}}{\frac{RSS}{n-k-1} \rightarrow \text{avg. unexplained var per df.}}$$

If value of F-stat $\rightarrow$ ① Very High

It means ESS is high, thus regression line is working.

$\therefore$ F-stat value = High, thus p-value is very low.

p-value $< \alpha$

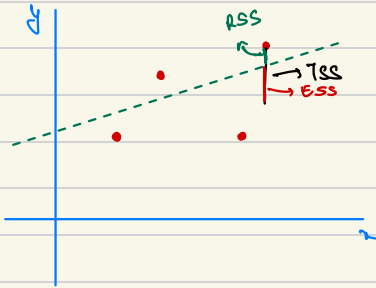
H_0 : Reject.

② Very low : $RSS \uparrow$, $F\text{-stat} = \downarrow$, $p\text{-value} = \uparrow$ $\therefore$ p-value $> \alpha$
 $\therefore H_0$: Accept.

Strength of the relationship :

R^2 (R Squared) :

Goodness of fit.



$$R^2 = \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS}$$

$$R^2 = 1 - \frac{RSS}{TSS}$$

R^2 is a proportion : Value b/w (0 to 1).

ESS $\xrightarrow[\text{full}]{\text{max}}$ TSS.

$R^2 = \frac{TSS}{TSS} = 1$: Does not explain any of variance.

$R^2 = 0$: Model explain all variance.

Disadvantage of R^2 : R^2 can be misleading sometimes. If lot of ilp var. & y, then R^2 is not good metric.

x_1	y	$r^2_{\text{score}} = 0.4$
cgpa	iz	
x_1	x_2	y
		$r^2_{\text{score}} = 0.6$

$$x_2 \rightarrow y$$

or $x_2 \nrightarrow y$

But if $x_2 = \text{temp.}$

$$r^2_{\text{score}} = 0.4$$

No decrease in R^2 score.

Adjusted R^2 Score :

$$R^2 = 1 - \left[\frac{(1 - R^2) * (n - 1)}{(n - k - 1)} \right]$$

$R^2 \rightarrow R^2$ score.

$n \rightarrow$ no. of observations

$k \rightarrow$ no. of predictor variables.

Which one to choose ?

Model comparison $\rightarrow$ adj R^2

Model interpretation $\rightarrow R^2$

Model selection / overfitting $\rightarrow$ adj R^2

Red flag $\rightarrow$ Big quantities.

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```

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```

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Confidence Interval.

Relation b/w

$x_1, x_2, \dots, x_n \rightarrow y$

$x_1 \rightarrow TV$ $x_2 \rightarrow Radio$ $x_3 \rightarrow Newspaper$ $y \rightarrow Sales$.

T-Statistic :

Simple L.R.

$x | y \rightarrow \beta_0, \beta_1$
 $\uparrow$ intercept $\uparrow$ coef

1. for slope $\Rightarrow H_0 : \beta_1 = 0$ (i.e. No relation b/w x & y)
 $H_1 : \beta_1 \neq 0$

for intercept $\Rightarrow H_0 : \beta_0 = 0$ (i.e. regression line passes through origin.)
 $H_1 : \beta_0 \neq 0$

2. Estimate the slope & intercept. (Apply LR to find b_0, b_1)

$$t\text{-statistic}_{b_1} = \frac{b_1 - 0}{SE(b_1)}$$

$$t\text{-statistic}_{b_0} = \frac{b_0 - 0}{SE(b_0)}$$

3. $SE(b_1) = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{(n-2) \sum (x_i - \bar{x})^2}}$

$$SE(b_0) = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{(n-2)} \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right]}$$

4. T-Score

df

α

Two-tailed

Get p-value

5. $p\text{-val} < \alpha = 0.05$

6. H_0 : Rejected i.e. ($\beta_1 \neq 0$)

$\therefore$ Relation b/w x_i & y_i .

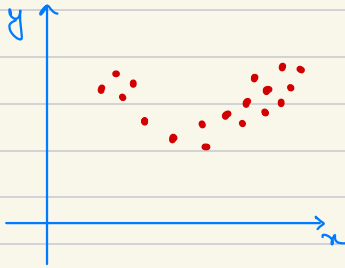
How confidence interval is calculated?

$$C.I_{b_0} = b_0 \pm t\text{-value} * S.E(b_0)$$

$$C.I_{b_1} = b_1 \pm t\text{-value} * S.E(b_1)$$

These C.I represents the range within which the true population regression coeff. are likely to fall with a specified level of confidence (eg., 95%).

Polynomial regression



$$y = mx + c$$

$$y_0 = \beta_0 + \beta_1 x$$

→ Simple Linear Regression.

$$y_0 = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

→ Multiple Linear Regression.

In above data, it's impossible to fit a linear line, thus we use polynomial regression.

$x \mid y$
↓
extract polynomial features.
 $x^0 \quad x^1 \quad x^2 \quad (\text{degree}=2)$

Q1) $x \mid y$
 $(35, 100) \xrightarrow{\text{Slope}} (1 \times 2)$
↓
 $x^0 \quad x^1 \quad x^2 \quad , y$
 $35^0 \quad 35^1 \quad 35^2 \quad , y$
 $1 \quad 35 \quad 1225 \quad , y \xrightarrow{\text{Slope}} (1 \times 4)$
↓
Training

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 \quad (\text{for degree}=2)$$

How to set value of degree?

degree $\rightarrow$ Hyperparameter.

if degree $\downarrow \rightarrow$ graph underfit

if degree $\uparrow \rightarrow$ it tries to touch every point (i.e. overfit)

x_1, x_2, y

for degree = 2

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \beta_3 x_3^2 + \dots + \beta_n x_n^2$$

$\hookrightarrow$ Polynomial regression.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2^2 + \hat{\beta}_3 x_3^2 + \dots + \hat{\beta}_n x_n^2$$

We find value of $\hat{y}$, $\hat{\beta}_0$, $\hat{\beta}_1$ --- etc. $\therefore$ It's categorized under L.R.

Assumptions of linear regression

☆

Assumptions of LR :

1. Linear.

- ## 2. Normality of Residuals.

- ### 3. Homoscedasticity.

- #### 4. No Autocorrelation

5. No or little Multicollinearity.

1. What

2. What if it fails.

- ### 3. Deck

- #### 4. Remedy.

1. Linear :

There is a linear relationship b/w x (independent variable) and y (dependent var.)

What happens if assumption is violated?

1. Bias in parameter estimates.

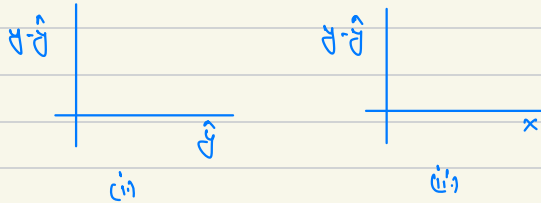
2. Reduce predictive accuracy.

3. Invalid hypothesis test and Confidence interval.

How to check this assumption?

1. Scatter plot. $\rightarrow (x_1, y), (x_2, y)$ graphs.

2. Residual plot $\rightarrow \{r(x, y) \xrightarrow{\text{Calc.}} \hat{y} \rightarrow (y - \hat{y})$
 $\hookrightarrow$ residual



Residual Plot

if linear $\rightarrow$ Residual plot random around 0.

3. Polynomial regression.

Find R^2 score, if R^2 score improves then non-linear.

↓
(polynomial reg. vs LR)

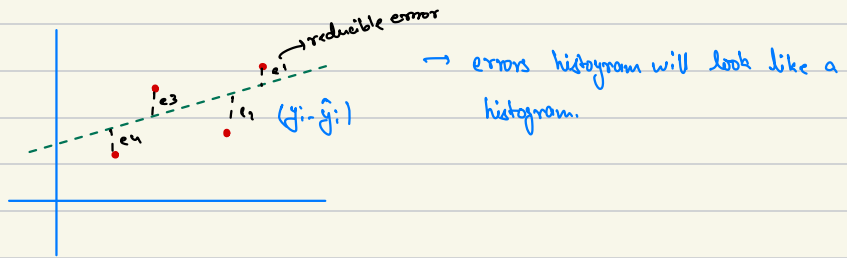
What to do when the assumption fails?

1. Transformation: Apply math. transformation on column. (eg. log, box-cox etc)
- jee-johnson (best)
2. Polynomial regression
3. Piecewise regression
4. Non-parametric / semi-parametric regression: Switch to GAM, kernel, splines etc.

↑
Generalized Additive Method.

2. Normality of Residual:

The error term are assumed to follow a normal distribution with the mean of zero and a const. variance.



$$e_i \sim N(0, \sigma^2)$$

What happens if this assumption is violated?

1. Inaccurate hypothesis test.
2. Invalid confidence interval.
3. Model performance.

How to check this assumption?

1. Histogram of residuals. $\rightarrow x, y \rightarrow \hat{y} \rightarrow y - \hat{y} \rightarrow \text{Histogram/ kde.}$
 2. Q-Q plot.
 3. Statistical Test. $\rightarrow$ like Shapiro wilk test, Omnibus test, Jarque-Bera test.
 4. Residual plot.
- Test of normality of residuals.

Omnibus Test.

1. Decide H_0 : Residuals are Normally distributed.
 H_1 : Not normally distributed.
2. Fit LR model. $\rightarrow \text{coef} \rightarrow \hat{y}$
3. Calc. the residuals $(y - \hat{y})$
4. Calc. the skewness.
5. Calc. the kurtosis.
6. Calc. the Omnibus test statistic.

$$K^2 = n \left[\frac{(\text{Skewness})^2}{6} + \frac{(\text{Kurtosis})^2}{24} \right] \quad \left(\text{Chi-sq. distribution df} = 2 \right)$$

$n \rightarrow$ no. of observations.

7. Determine p-value.
8. Compare p-value & α .

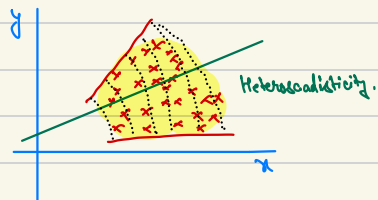
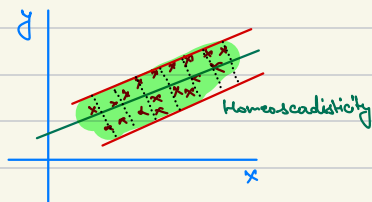
What to do when the assumption fails?

1. Model selection techniques $\rightarrow$ like Feature Selection.
2. Robust regression. $\rightarrow$ LMS, LTS etc. or Transformation.
3. Non-parametric or semi-parametric methods $\rightarrow$ Kernel techniques, GAM.
4. Use bootstrapping.

If $n > 30$, distribution is normal due to Central Limit Theorem (CLT).

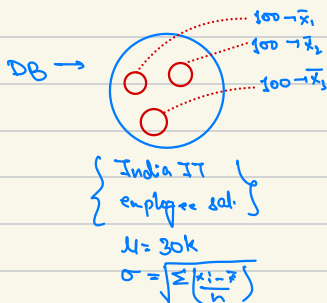
3. Homoscedasticity.

The spread of the error term should be constant across all levels of independent variable (x). If the spread of residuals changes systematically it leads to heteroscedasticity which can affect the efficiency of the estimates.



If homoscedasticity is absent, then, S.E of $(b_0, b_1, \dots, b_n)$ is unreliable.

Standard Error.



(Std of the sample means is standard error.)

$$S.E = \frac{\sigma}{\sqrt{n}}$$

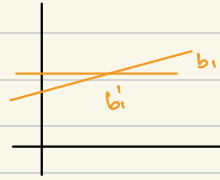
$\sigma \rightarrow$ pop. std.

$n \rightarrow$ observations.

$$N(\mu, \sigma/\sqrt{n})$$

Sol 1 exp

- - $\rightarrow$ 100 $SE(b_0)$ $SE(b_1)$
 - -
 - -



$$b_{1(1)}, b_{1(2)}, \dots$$

$\downarrow$
 Std. dev $\rightarrow$ $SE(b)$
 (precision)

What happens when this assumption is violated?

1. Inefficient estimates.
2. Inaccurate hypothesis test.
3. Invalid confidence interval.

What to do when assumption fails?

1. Transformations
2. Weighted Least Squares (WLS)

(To be Continued...)

Multicollinearity

It's a statistical phenomena that occurs when two or more independent variables in a multiple regression model are highly correlated. In other words, these vars. exhibit a strong linear relationship, making it difficult to isolate the individual effects of each var. on the dependent variable.

$$n=100. \quad \begin{array}{c|cc} \text{gpa} & i_j & lpa \\ \hline 8 & 80 & 8 \\ - & - & - \\ - & - & - \end{array}$$

If 2 input col like gpa & i_j have correlation (i.e. linear relationship), then system have multicollinearity.

Like, if $i_j \uparrow$ then gpa is high. } multicollinearity.
if $i_j \uparrow$ and backlogs $\downarrow$

Without correlation. like gpa & date-of-birth.

$$lpa \rightarrow \beta_0 + \beta_1 \text{gpa} + \beta_2 i_j$$

Coefficient are weightage.

In multicollinearity if we increase value of gpa the value of i_j will also increase, thus, the overall lpa increases, & the value of β_1 / β_2 is unreliable.

$\beta_0, \beta_1, \beta_2 \rightarrow$ unreliable.

When is Multicollinearity bad?

1. Inference : Understanding the relationship b/w variables of the model.
2. Prediction : Estimate the forecast for new unseen data.

Multicollinearity does not effect if we do prediction but it effects when we do inferences.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

└───┘
correlated.

then $\beta_1, \beta_2 \rightarrow$ unreliable.

because if $x_1 \uparrow$ then $x_2 \uparrow$ vice versa.

we cannot give relation b/w y & x_1, y & x_2

Thus no inference possible.

for Prediction if x_1 & x_2 are correlated.

$$\text{then, } x_1 = a_0 + a_1 x_2 + \eta \rightarrow \text{error}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$y = \beta_0 + \beta_1 (a_0 + a_1 x_2 + \eta) + \beta_2 x_2 + \varepsilon$$

$$y = \beta_0 + \beta_1 a_0 + \beta_1 a_1 x_2 + \beta_1 \eta + \beta_2 x_2 + \varepsilon$$

$$y = (\beta_0 + \beta_1 a_0) + \beta_1 a_1 x_2 + \beta_2 x_2 + (\beta_1 \eta + \varepsilon)$$

└─┘
 c

└─┘
 r (new error)

Optimization
of this model $\rightarrow$

$$\boxed{y = c + \beta_1 a_1 x_2 + \beta_2 x_2 + r} \quad (\text{L.R. eqn. b/w } y \text{ \& } x_2)$$

$\therefore$ Prediction capability remain same as still

L.R is being applied. / or the new gen. also
does L.R.

Multicollinearity (Mathematical) :



What exactly happens in Multicollinearity (Mathematics) ?

1. Difficulty in identifying the most important predictors., it becomes challenging to determine which var. has the most significant impact of y .
2. Inflated Standard Errors. (i.e. $S.E(\beta)$ takes very high value)
3. Unstable & unreliable estimates (i.e. β_1, β_2 becomes unreliable)

Perfect Multicollinearity :

Linear correlation b/w vars.

$$\text{Correlation} \rightarrow x_1 = a_1 x_2 + a_0 + \text{error}$$

$$\text{Perfect correlation} \rightarrow x_1 = a_1 x_2 + a_0$$

i.e. If you have a_1, a_0 & x_2 , you can exactly calculate value of x_1 .

$$\text{eg. } \begin{array}{ccc} \text{cgpa} & | & \text{percent} & | & \text{lpa} \\ \hline & & & & \end{array}$$

then, $\text{percent} = 30 \times \text{cgpa} + 0$ ^{No-error} : perfect collinearity.

$$\text{Imperfect multicollinearity : } x_1 = a_1 x_2 + a_0 + \text{error}$$

Generally we have imperfect multicollinear dataset in real world.

We are talking about perfectly multicollinear data :—

cgpa		percent		lpa
8		80		3
6		60		4

We have to find all the β values (i.e. $\beta_0, \beta_1, \beta_2$)

$$l_{pa} = \beta_0 + \beta_1 gpa + \beta_2 percent + error.$$

Finding β with OLS technique.

$$\beta = (x^T x)^{-1} x^T y.$$

Design matrix

$$\begin{bmatrix} 1 & 8 & 80 \\ 1 & 6 & 60 \end{bmatrix}$$

$$\beta = (x^T x)^{-1} x^T y$$

$$\begin{bmatrix} 1 & 1 \\ 8 & 6 \\ 80 & 60 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 8 & 80 \\ 1 & 6 & 60 \end{bmatrix}_{2 \times 3}$$

$$\begin{bmatrix} 2 & 14 & 140 \\ 14 & 8 \times 8 + 6 \times 6 & 8 \times 80 + 6 \times 60 \\ 140 & 8 \times 8 + 6 \times 6 & 8 \times 80 + 6 \times 60 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} 2 & 14 & 140 \\ 14 & 100 & 1000 \\ 140 & 1000 & 10^4 \end{bmatrix}_{3 \times 3}$$

$$\det(x^T x) \Rightarrow 2 \times (10^6 - 10^6) - 14(10) + 140(0)$$

$$\det(x^T x) \Rightarrow 0$$

It means it's a singular matrix

Thus, we cannot find inverse.

If we cannot find inverse, so we cannot find β .

Non-Perfect Multicollinearity. :

$$\beta = (X^T \cdot X)^{-1} \cdot X^T \cdot Y$$

$$\text{var}(\beta) = \text{S.E}(\beta) = \sigma^2 (X^T \cdot X)^{-1}$$

$$\text{S.E}(\beta_0) \quad \text{S.E}(\beta_1) \quad \text{S.E}(\beta_2)$$

cgpa | iq | lpa $\rightarrow$ sample data. (n=100)

$$lpa = \beta_0 + \beta_1 \text{cgpa} + \beta_2 \text{iq}$$

diff. β_0 β_1 β_2 for 10 datasets

$\therefore$ We take S.E(β_0)

$$\text{var}(\beta) = \underset{\substack{\uparrow \\ \text{const}}}{\sigma^2} (X^T \cdot X)^{-1}$$

$$= \sigma^2 \begin{bmatrix} a & - & - \\ - & b & - \\ - & - & c \end{bmatrix}$$

variance-covariance Matrix.

$a \rightarrow \text{var}(\beta_0)$ $b \rightarrow \text{var}(\beta_1)$ $c \rightarrow \text{var}(\beta_2)$

$$\text{S.E}(a) = \sqrt{\text{var}(\beta_0)}, \quad \text{S.E}(b) = \sqrt{\text{var}(\beta_1)}, \quad \text{S.E}(c) = \sqrt{\text{var}(\beta_2)}$$

$$\text{S.E}(\beta) = \sqrt{\text{diagonal}(\sigma^2 (X^T X)^{-1})}$$

$$\beta = (X^T \cdot X)^{-1} \cdot X^T \cdot Y$$

Perfect Multicollinearity : det = 0

Strong Multicollinearity : det = very small.

$$\text{inverse} \rightarrow \frac{1}{\text{det.}} [\text{adj. Matrix}]$$

So, if det. very small

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \rightarrow \text{values will inflate.}$$

$\therefore$ S.E will be high.

Types of Multicollinearity :

1. Structural multicollinearity : Arises due to way which the variables are defined or the model is constructed. (e.g. Perfect multicollinearity)
Ottor Encoding
2. Data-Driven multicollinearity : Naturally present in the data.

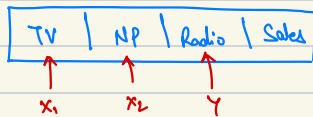
How to detect multicollinearity ?

1. Correlation
2. VIF (Variance Inflation Factor)
3. Condition Number

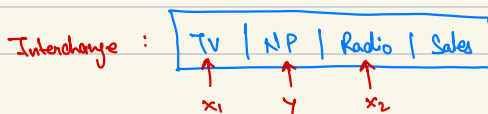
Correlation : Is there linear relationship b/w x_1, x_2 .
pandas corr() fn. $\rightarrow$ df.corr()

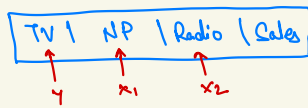
High Correlation $\rightarrow$ Value > 0.8
Value < -0.8

Variance Inflation Factor (VIF) :



Makes Radio as output.





Run I.R for all, find R^2 score.

$$VIF_i = \frac{1}{1 - R_i^2}$$

if $VIF_i > 5$ or 10 : There is multicollinearity.

Condition Number : Calc. as the ratio of largest eigen value vs smallest eigen value.

It tells ill conditioning of the matrix.

Tells how $\det(X) \approx 0$

if significant no. > 30 : Multicollinearity.

How to remove multicollinearity?

1. Collect more data.
2. Remove one of the highly correlated variable.
3. Combine correlated variable.
4. Use Partial Least Squares Regressions (PLS). $\rightarrow$ (PCA then LR)

